Assume the following throughout this entire sheet

- $\mathbf{F}(x,y,z) = [M(x,y,z), N(x,y,z), P(x,y,z)]$ a vector-valued function in $\mathbf{R}^3$ with continuous partial derivatives.
- $\mathbf{G}(x,y) = [M(x,y), N(x,y)]$ a vector-valued function in $\mathbf{R}^2$ with continuous partial derivatives.
- $g(x,y,z)$ a scalar-valued function with continuous first partial derivatives
- $\mathbf{r}(t)$ represents some parameterization of a curve C for $a \leq t \leq b$
- C_0 is a simple closed curve enclosing a region R

Operators: $\nabla g(x,y,z) = [g_x, g_y, g_z]$ div $\mathbf{F} = M_x + N_y + P_z$ curl $\mathbf{F} = [P_y - N_z, M_z - P_x, N_x - M_y]$

Line Integrals: Let $r(t)$ be a parameterization for the curve C

Vector-Valued: $$\int_C \mathbf{F}\cdot d\mathbf{r} = \int_a^b \mathbf{F}(\mathbf{r}(t))\cdot \mathbf{r}'(t)\,dt$$ *Scalar-Valued*: $$\int_C g\,ds = \int_a^b g(\mathbf{r})\cdot|\mathbf{r}'(t)|dt$$

Surface Integrals: Let $r(u,v)$ be a parameterization for the surface S

Vector-Valued: $$\iint_S \mathbf{F}\cdot\mathbf{n}\,dS = \iint_R \mathbf{F}(\mathbf{r}(u,v))\cdot\mathbf{N}\,du\,dv$$ *Scalar-Valued*: $$\iint_R g(\mathbf{r}(u,v))\cdot|\mathbf{N}|dudv$$

Stokes' Theorem: Let C be a closed curve which is the boundary for a surface S. Then

$$\oint_C \mathbf{F}\cdot d\mathbf{r} = \iint_S (\text{curl}\,\mathbf{F})\cdot\mathbf{n}\,dS$$

* Recall: Green's Theorem is a special case of Stokes' Theorem where S is in the xy-plane and $\mathbf{n} = [0,0,1]$.

Divergence Theorem: Let S be a surface that is the boundary of T. Then

$$\iint_S \mathbf{F}\cdot d\mathbf{S} = \iiint_T \text{div}\,\mathbf{F}\,dV$$

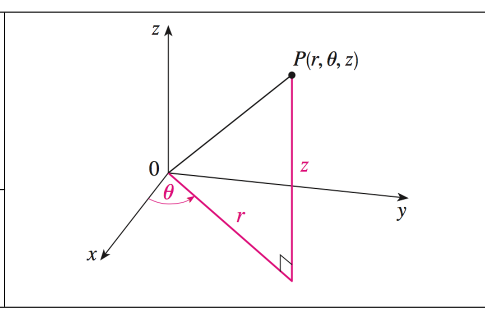
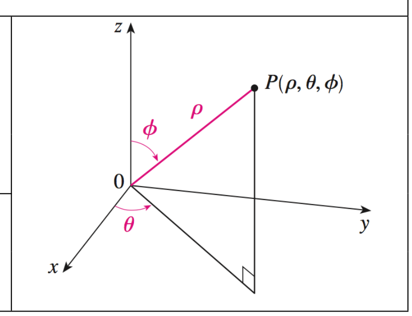
Complex Numbers:

Let $z = re^{\theta i} = r(\cos\theta + i\sin\theta)$

$$z^n = r^n e^{n\theta i} = r^n\left[\cos(n\theta) + i\sin(n\theta)\right]$$

$$z^{1/n} = r^{1/n}e^{\frac{\theta+2\pi k}{n}i} = r^{1/n}\left[\cos\left(\frac{\theta+2\pi k}{n}\right) + i\sin\left(\frac{\theta+2\pi k}{n}\right)\right] \quad \text{for } k = 0,1,2,\ldots,n-1$$

Special Coordinates

Cylindrical		Spherical	
$x = r\cos\theta$ $y = r\sin\theta$ $z = z$		$x = \rho\sin\phi\cos\theta$ $y = \rho\sin\phi\sin\theta$ $z = \rho\cos\phi$	
$x^2 + y^2 = r^2$ $dV = r\,dr\,d\theta\,dz$		$x^2 + y^2 + z^2 = \rho^2$ $dV = \rho^2\sin\phi\,d\rho\,d\theta\,d\phi$	

Fourier Series

$$f(x) = \sum_{n=-\infty}^{\infty} d_n e^{i\pi n x/L} \qquad \text{where} \qquad d_n = \frac{1}{2L} \int_{-L}^{L} f(x)\, e^{-i\pi n x/L}\, dx$$

$$f(x) \text{ is even:} \quad d_n = \frac{1}{L} \int_{0}^{L} f(x) \cos\left(\frac{\pi n x}{L}\right)\, dx$$

$$f(x) \text{ is odd:} \quad d_n = \frac{-i}{L} \int_{0}^{L} f(x) \sin\left(\frac{\pi n x}{L}\right)\, dx$$